RAIN ENHANCEMENT TECHNOLOGIES

OPERATIONS REPORT: LA SAL, UT (WA25001)

January 2026

Version 01

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Revision History

Ver #	Comments	Author	Date
0.0c	Original Document Development – Draft	Jeffrey Chagnon; Rutuja Dongre; Scott Morris	01/13/2026
1.0	Released for External Review	Jeffrey Chagnon; Rutuja Dongre; Scott Morris	01/14/2026

2 *Executive Summary*

This report provides a preliminary review of the Weather Enhancement Technology Array (WETA) installed by Rain Enhancement Technologies (RET) on the Flat Iron Mesa, outside Moab, Utah. WETA is a self-sufficient and "off grid" solution installed to provide enhancement of precipitation in year-round operations (snow and rainfall) over the La Sal Mountain Ranges. The focus of this report is on the December 2025 operating period.

2.1 Operational Summary

1. This report covers 31 days from 12/01/2025 to 12/31/2025.
   a. WETA operated on thirteen days in December 2025.
   b. Precipitation was observed on twelve of those days.
   c. Operations were suspended from 12/23 through 12/25 to avoid significant rainfall on snowpack at high altitude.
2. December 2025 was a climatologically dry month, despite the occurrence of precipitation on twelve days.
3. Relatively warm and dry conditions, coupled with the rain event on 12/23-12/25, resulted in a reduction of snowpack during the month. Some of this loss has already been recuperated in January.
4. Despite the month being relatively dry, the evidence presented in this report supports the likelihood that precipitation was enhanced by WETA over the La Sal range.
5. Radiometrics radiometer data is being processed. Preliminary analysis indicates orographic enhancement of vertically-integrated liquid water over the La Sal range.
6. Power issues related to the radiomenter elaborated in the previous report are ongoing.
   a. Changes implemented remotely to "work around" until full repair implemented.
   b. Generator installation expected early February 2026.

2.2 Actions and Improvements

1. December precipitation and snow pack depth was evaluated against a historical baseline.
2. Addition of load shedding and backup generator for improved power reliability on site (planned for Jan / Feb 2026).
3. Installation of further instrumentation in La Sal Ranges (planned Mar/Apr 2026).
4. For each event, identification of treatment and control probabilities based on HySPLIT simulations (next reporting cycle).
5. WETA operations will continue to target predicted precipitation events that are not considered adverse weather conditions or impacting snow pack.

3 *Operations Summary*

Figure 1 details the operating schedule during the reporting period. WETA was operated during periods when forecasts indicated any chance of precipitation over the target area, including marginal cases involving scattered light showers. Operations were suspended from 23 to 25 December due to the likelihood of moderate rainfall over high-altitude snowpack.

DATE(S)	**WETA on/off**
1-Dec	0134 off
2-Dec	1158 on / 2356 off
3-Dec to 5-Dec	off
6-Dec	254 on / 1740 off
7-Dec to 15-Dec	off
16-Dec	2135 on
17-Dec	on
18-Dec	1206 off
19-Dec	off
20-Dec	0208 on
21-Dec	on
22-Dec	0100 off
23-Dec	1310 on / 1530 off
24-Dec to 25-Dec	off
26-Dec	1403 on
27-Dec	on
28-Dec	0508 off
29-Dec to 31-Dec	off

Figure 1: WETA operating schedule during December 2025. Green shading indicates periods of operation.

4 *Data and Analysis*

4.1 Summary of Precipitation Events

December 2025 was a relatively dry month. Figure 2 presents the daily time-series of precipitation measured at weather stations and SNOTEL in the region. Intermittent periods of light showers occurred during the first three weeks of December. WETA was operational during all but two of these marginal events. The most significant period of precipitation began on 24 December and terminated on 28th December. WETA was operational for all but the beginning of this event. The late start was due to concern about rain over snowpack.

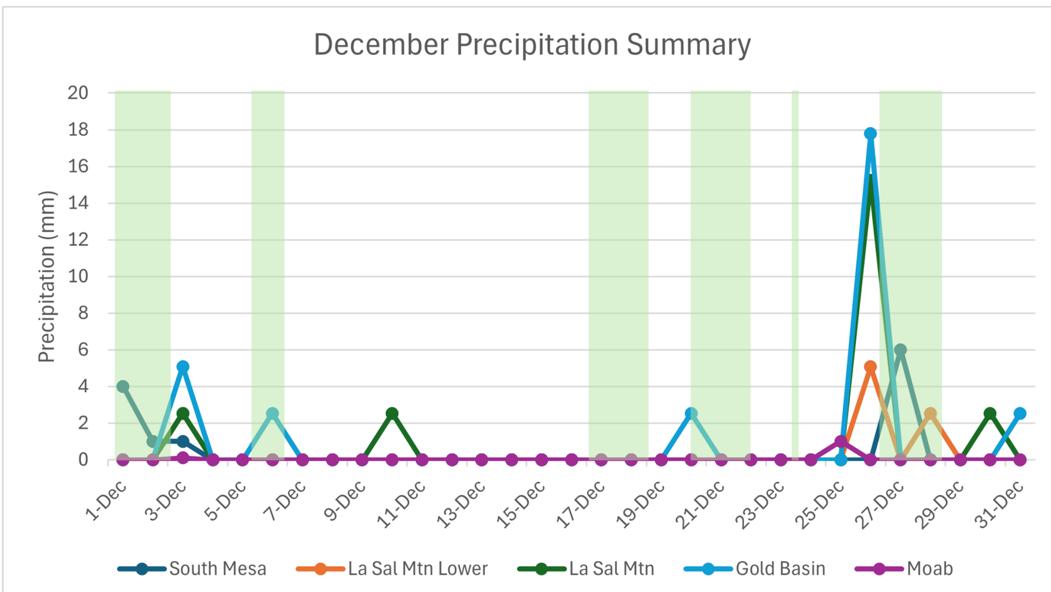

Figure 2: Summary of daily accumulated precipitation at reporting weather and SNOTEL stations.

4.2 Evaluation of Snow Pack Enhancement

Snow pack enhancement was evident in the target area. Climatological snow-depth baselines have been calculated at SNOTEL sites inside the target area as well as outside (but nearby). Figure 3 presents a comparison of two such sites: La Sal Mountain and Camp Jackson. The SNOTEL site at La Sal Mountain is at an elevation of 9580 ft and is inside the treatment area. Camp Jackson is located at 8840 ft in the Abajo Mountains to the south and outside of the treatment area. Both sites are at a similar altitude, receive similar annual snowfall, and are subject to a similar synoptic environment. The comparison presented in Figure 3 reveals the following:

1. In a typical year, snow pack depth in the La Sal range is similar to that in the Abajo mountains.
2. Both sites had a much shallower snowpack than average in December 2025 (see red circles in Fig.3).
3. Despite the dry conditions, La Sal maintained a deeper snowpack than Camp Jackson. The difference was above the climatological mean.

4.3 Analysis of Radiometer Data

Radiometer-estimated liquid water content indicated a significant increase in cloud liquid water over La Sal during the active period at the end of December 2025. Fig. 4 presents the difference in vertically-integrated liquid water between two azimuthal directions – one to the northeast directed towards the La Sal range, and one directed vertically. The higher values over the La Sal range were evident both before and after December 26th when WETA was suspended and operating, respectively. A longer record of analysis will indicate whether WETA operations amplify this apparent orographic effect.

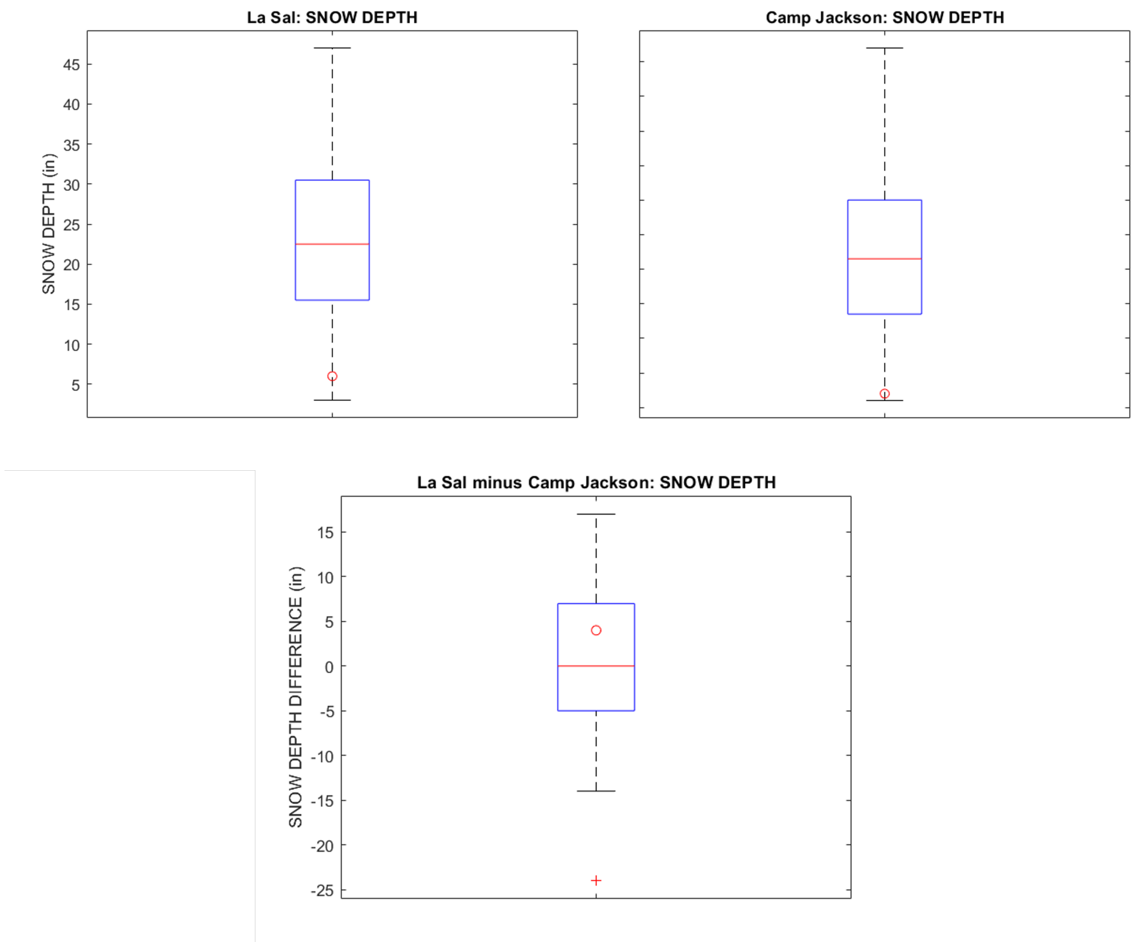

Figure 3: Box and whisker plots demonstrating SNOTEL-measured climatological December snow depth at (top left) La Sal Mountain, and (top right) Camp Jackson. The difference in monthly precipitation is shown in the bottom panel. The red circle in each panel indicates values for December 2025.

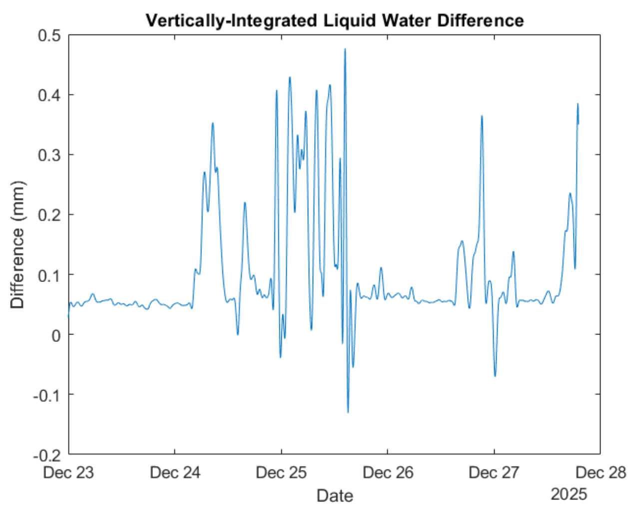

Figure 4: Low-pass filtered difference in radiometer-estimated vertically-integrated liquid water between two azimuthal directions: northeast (treatment) minus zenith (control).

5 *References*

5.1 Data Sources

Type of Data Source	Specific Locations	Data Description	Granularity/ Frequency
Radar	Grand Junction, CO	Radar reflectivity, velocity, precipitation accumulation, and all Level II and III fields at all elevation angles	$\sim$5 minutes
Radiometer	Moab, UT	Relative humidity, liquid water content in all scan directions	One minute
Weather Station	South Mesa, UT; Gold Basin, UT; Moab, UT; US-191 at MP 104 Flat Iron, UT; Hole N The Rock, UT; La Sal, UT; SR-46 at MP 12.5 La Sal Divide (UT DOT), UT	Weather variables such as precipitation, wind speed, wind direction, temperature, etc.	Hourly
SNOTEL	Castle Valley, UT; Gold Basin, UT; La Sal Mtn, UT; La Sal Mtn Lower, UT	Snow / water equivalent monitoring	Daily